Astronomy A1: Solutions

The Night Sky and Celestial Motion · MAIS PAC

Problem 1.

(a) Using $\theta = 2\mathbf{R}/\mathbf{D}$ in radians and multiplying by $(180/\pi) \times 60' = 3437.75'/\text{rad}$:

$$\theta_{\odot} \approx (2(6.96 \times 10^5))/(1.50 \times 10^8) \times 3437.75' \approx 31.9', \; \theta_{\mathbf{M}} \approx (2(1{,}737))/(3.84 \times 10^5) \times 3437.75' \approx 31.1'. \tag{1}$$

Although the Sun is ~400 times larger in radius than the Moon, it is also ~390 times farther away, making both disks span roughly half a degree (~31′–32′) on our sky!

(b) Near lunar perigee (closest approach), $\theta_{\mathbf{M}}$ increases to $\sim 33.5' > \theta_{\odot}$, completely covering the solar photosphere and revealing the faint solar corona (a **total** eclipse). Near apogee, $\theta_{\mathbf{M}}$ shrinks to $\sim 29.4' < \theta_{\odot}$, leaving a bright ring of sunlight around the Moon's silhouette (an **annular** eclipse). Eclipses do not happen every month because the Moon's orbital plane is tilted by 5.1° relative to the ecliptic, so at most new moons the Moon passes several degrees north or south of the Sun; eclipses occur only when new moon coincides with one of the two orbital nodes.

Problem 2.

(a) At $\varphi = 47.9°$ N, $90° - \varphi = 42.1°$. Stars within 42.1° of the north celestial pole—that is, $\boldsymbol{\delta > +42.1°}$—are circumpolar and never set. Symmetrically, stars within 42.1° of the south celestial pole—$\boldsymbol{\delta < -42.1°}$—never rise above Ulaanbaatar's horizon.

(b) Checking each star's declination:

- **Dubhe** ($\delta = +61.8° > +42.1°$): **circumpolar**. Its polar distance from the north celestial pole (at altitude 47.9°) is $90° - 61.8° = 28.2°$, so Dubhe reaches upper culmination at $\mathbf{h}_{\text{max}} = 47.9° + 28.2° = \mathbf{76.1°}$ and dips at lower culmination to $\mathbf{h}_{\text{min}} = 47.9° - 28.2° = \mathbf{19.7°}$ due north.
- **Vega** ($\delta = +38.8°$): **seasonal** (just 3.3° shy of being circumpolar, rising high to $\mathbf{h}_{\text{max}} = 80.9°$ in summer nights).
- **Sirius** ($\delta = -16.7°$): **seasonal** winter star, culminating due south at $\mathbf{h}_{\text{max}} = 90° - (47.9° - (-16.7°)) = \mathbf{25.4°}$ above the southern horizon.
- **Alpha Centauri** ($\delta = -60.8° < -42.1°$): **never visible** from Ulaanbaatar (peaking 18.7° below the southern horizon).

Problem 3.

(a) Imagine Earth froze its spin relative to the distant stars while completing one revolution around the Sun: an observer on Earth would still experience one apparent solar cycle (in reverse!). Because Earth spins in the **same** prograde direction as its orbit, one orbital revolution subtracts one solar day relative to the number of sidereal turns, so $\mathbf{N}_{\text{sid}} = \mathbf{N} + 1 = 366.2422$. Hence

$$\mathbf{T}_{\text{sid}} = (365.2422)/\,(366.2422) \times 24\ \text{h} \approx 23.93447\ \text{h} = 23\ \text{h}\ 56\ \text{m}\ 4.1\ \text{s}. \tag{2}$$

(b) Every solar day, any given star crosses the meridian 24 h $-$ $\mathbf{T}_{\text{sid}} \approx 3$ m 55.9 s earlier on our wristwatch. Over 30 days this accumulates to 30 × (24 h / 365.24) ≈ 1.97 h ≈ **1 h 58 m** earlier, so Orion will transit at roughly **21:02** local time.

Problem 4.

(a) The faster inner body advances in angle at $\omega_1 = 2\pi/\mathbf{P}_1$ while the slower outer body advances at $\omega_2 = 2\pi/\mathbf{P}_2$. Between two consecutive alignments, the inner body gains one full lap (2π radians) on the outer body: $(\omega_1 - \omega_2)\mathbf{S} = 2\pi$. Dividing through by $2\pi\mathbf{S}$ gives $(1)/\,(\mathbf{S}) = (1)/\,(\mathbf{P}_1) - (1)/\,(\mathbf{P}_2)$.

(b) Relative to the Earth–Sun line (which rotates at $1/\mathbf{P}_{\oplus}$), the Moon's phase cycle satisfies

$$(1)/\,(\mathbf{S}) = (1)/\,(27.32\ \text{d}) - (1)/\,(365.25\ \text{d}) \Rightarrow \mathbf{S} \approx 29.53\ \text{days}. \tag{3}$$

(c) Neptune crawls along its orbit so slowly ($1/165\ \text{yr}^{-1}$) that after Earth completes one 365.25-day lap, Neptune has barely moved ~2.2°, taking Earth only 2.2 days to catch up! Mars, by contrast, moves more than half as fast as Earth, leading Earth on a long chase over more than two full years (~780 days) before Earth laps it again.